# Marking guide

Year 10 Mathematics Yearly Test 2026 · 55 marks · 1 hour 15 minutes. Answers for every question are below, and the test pages that follow have the answers written in red.

| Q | Topic and where it comes from | Answer | Marks | Marking notes |
|---|---|---|---|---|
| **Section A · Financial mathematics · 15 marks · Chapter 1 review and booklet** | | | | |
| 1 | Percentage of an amount<br>Ch 1 review MC 1 · Booklet 1.1 | **B ($24.92)** | 1 | |
| 2 | GST<br>Ch 1 review MC 3 · Booklet 1.2 | **A ($5.00)** | 1 | |
| 3 | Discount<br>Ch 1 review MC 7 · Booklet 1.3 | **D ($41.44)** | 1 | |
| 4 | Simple interest<br>Ch 1 review MC 9 · Booklet 1.6 | **C ($120.00)** | 1 | |
| 5 | Percentage increase and decrease<br>Ch 1 review SA 2, 3 · Booklet 1.2 | **(a) $1 084.16**<br>**(b) $4 557.00** | 2 | |
| 6 | Mark-up and profit<br>Ch 1 review SA 4 · Booklet 1.3 | **(a) $11.40**<br>**(b) $3.80** | 2 | (b) Follow-through from (a): accept selling price − 7.60. |
| 7 | Wages and overtime<br>Ch 1 review MC 4, 6 · Booklet 1.4 | **(a) $473.40**<br>**(b) $94.68** | 2 | |
| 8 | Commission<br>Ch 1 review SA 12 · Booklet 1.5 | **$35.55** | 1 | |
| 9 | Gross and net pay<br>Ch 1 review SA 10 · Booklet 1.5 | **(a) $1 040.04**<br>**(b) $874.95** | 2 | (b) Follow-through: accept their (a) − 165.09. |
| 10 | Compound interest<br>Ch 1 review ER 1 · Booklet 1.7 | **(a) $5 624.32**<br>**(b) $624.32** | 2 | (b) Follow-through: accept their (a) − 5000. Accept $5624.32 rounded to $5624. |
| **Section B · Measurement · 15 marks · Chapter 2 review and booklet** | | | | |
| 11 | Units of length<br>Ch 2 review MC 1 · Booklet 2.1 | **C (1 000)** | 1 | |
| 12 | Area of a rectangle<br>Ch 2 review MC 5 · Booklet 2.4 | **D (12 cm²)** | 1 | |
| 13 | Volume of a cube<br>Ch 2 review MC 9 · Booklet 2.7 | **B (27 cm³)** | 1 | |
| 14 | Scientific notation<br>Ch 2 review SA 8a | **A ($2.32 \times 10^5$)** | 1 | |
| 15 | Converting units<br>Ch 2 review SA 1a · Booklet 2.1 | **(a) 5 300 m**<br>**(b) 2 500 mL** | 2 | |
| 16 | Perimeter<br>Ch 2 review SA 3 · Booklet 2.2 | **38 m** | 1 | |
| 17 | Circumference and area of a circle<br>Ch 2 review SA 4 · Booklets 2.3, 2.5 | **(a) 18.8 m**<br>**(b) 28.3 m²** | 2 | Accept 2 d.p. (18.85, 28.27) or answers using $\pi \approx 3.14$ (18.84, 28.26). |
| 18 | Area of a triangle and a trapezium<br>Ch 2 review MC 6, SA 6 · Booklet 2.4 | **(a) 30 cm²**<br>**(b) 20 cm²** | 2 | |
| 19 | Surface area of a prism<br>Ch 2 review SA 7b · Booklet 2.6 | **40 cm², 16 cm²; surface area 136 cm²** | 2 | 1 mark: both rows correct (20, 40 and 8, 16). 1 mark: total 136 cm². |
| 20 | Volume and capacity of a cylinder<br>Ch 2 review MC 10, SA 9 · Booklet 2.7 | **(a) 75.4 cm³**<br>**(b) 75.4 mL** | 2 | Accept 75.40 or 75 (2 d.p. or whole). (b) Follow-through from (a). |

| Q | Topic and where it comes from | Answer | Marks | Marking notes |
|---|---|---|---|---|
| **Section C · Statistics and probability · 14 marks · Chapter 4 review and booklet** | | | | |
| **21** | Types of data<br>Ch 4 review MC 6 · Booklet 4.1 | **D (categorical and nominal)** | 1 | |
| **22** | Reading a stem-and-leaf plot<br>Ch 4 review MC 7 · Booklet 4.5 | **C (24)** | 1 | |
| **23** | Mode<br>Ch 4 review MC 8 · Booklets 4.4, 4.5 | **B (31)** | 1 | |
| **24** | Probability<br>Ch 4 review SA 1b | **C ($\frac{1}{2}$)** | 1 | |
| **25** | Mean, median and range<br>Ch 4 review MC 9, SA 8 · Booklet 4.4 | **(a) 6**<br>**(b) 5**<br>**(c) 12** | 3 | Order: 1, 3, 3, 5, 8, 9, 13. |
| **26** | Probability<br>Ch 4 review SA 2 | **(a) 2/8 = 1/4**<br>**(b) 3/8** | 2 | |
| **27** | Time-series graphs<br>Booklet 4.7 | **upward trend** | 1 | |
| **28** | Frequency tables, histograms and shape<br>Ch 4 review SA 6 · Booklets 4.2, 4.3, 4.6 | **(a) 20**<br>**(b) 45%**<br>**(c) see guide**<br>**(d) skewed** | 4 | (a) 20 (b) 9 ÷ 20 × 100 = 45% (c) heights 3 and 2, touching (d) skewed. |
| **Section D · Other Year 10 topics · 11 marks · Rest of the Year 10 syllabus** | | | | |
| **29** | Simplifying<br>Syllabus: Algebraic techniques | **A ($7x$)** | 1 | |
| **30** | Substitution<br>Syllabus: Algebraic techniques | **18** | 1 | |
| **31** | Solving equations<br>Syllabus: Equations | **(a) 6**<br>**(b) 5** | 2 | Accept any correct method, or the answer checked by substitution. |
| **32** | Table of values and graphing a line<br>Syllabus: Linear relationships | **y = 1, 3, 5, 7; points plotted and joined** | 3 | (a) 2 marks: all four correct; 1 mark: two or three correct. (b) Follow-through from their table. |
| **33** | Pythagoras' theorem<br>Syllabus: Pythagoras' theorem (Ch 2 review ER 2a) | **10 cm** | 2 | 1 mark: $6^2 + 8^2 = 100$ (or $c^2 = 100$). 1 mark: c = 10 cm. |
| **34** | Trigonometry: finding a side<br>Syllabus: Trigonometry | **5.6 cm** | 2 | 1 mark: x = 8 × tan 35°. 1 mark: 5.6 cm. (Radians mode gives −3.8: 1 mark for working.) |

**General marking notes**

- Give full marks for a correct answer, even with no working.
- Where a part uses an earlier answer, follow through: mark the method using the student's earlier answer.
- Do not take marks off for a missing $ sign or unit if the number is correct; note it for feedback.
- Rounding: accept 1 or 2 decimal places unless the question asks for a set number of places.

# Reporting

Each section is one content area, so each section total can be reported on its own.

| Section | Content area | Where it comes from | Marks | Share of test |
|---|---|---|---|---|
| A | **Financial mathematics**<br>Percentages, GST, profit, discount, wages, commission, interest | Chapter 1 review and booklet | 15 | 27% |
| B | **Measurement**<br>Units, perimeter, circles, area, surface area, volume | Chapter 2 review and booklet | 15 | 27% |
| C | **Statistics and probability**<br>Types of data, stem-and-leaf, mean, median, graphs, probability | Chapter 4 review and booklet | 14 | 25% |
| D | **Other Year 10 topics**<br>Algebra, equations, straight-line graphs, Pythagoras, trigonometry | Rest of the Year 10 syllabus | 11 | 20% |
| | **Chapter reviews and booklets (Sections A–C)** | Chapters 1, 2 and 4 | **44** | **80%** |
| | **Rest of the Year 10 syllabus (Section D)** | Algebraic techniques, equations, linear relationships, Pythagoras' theorem, trigonometry | **11** | **20%** |

## CLASS RECORD

| Student | A / 15 | B / 15 | C / 14 | D / 11 | Total / 55 | % |
|---|---|---|---|---|---|---|
| 1 | | | | | | |
| 2 | | | | | | |
| 3 | | | | | | |
| 4 | | | | | | |
| 5 | | | | | | |
| 6 | | | | | | |
| 7 | | | | | | |
| 8 | | | | | | |
| 9 | | | | | | |
| 10 | | | | | | |
| 11 | | | | | | |
| 12 | | | | | | |
| 13 | | | | | | |
| 14 | | | | | | |
| 15 | | | | | | |
| 16 | | | | | | |
| **Class average** | | | | | | |

SECTION A

# Financial mathematics

Questions **1–10** · **15 marks** · Suggested time **20 minutes**

____ / 15
Marker

Questions 1–4 are **multiple choice**. Circle the letter of the correct answer, like this: (B)

**1** What is 28% of $89? **1 mark**

(A) $28.00 (B) $24.92 (C) $64.08 (D) $113.92

**2** A price tag says **"$55 (GST inc.)"**. How much GST is included in the price? **1 mark**

(A) $5.00 (B) $5.50 (C) $50.00 (D) $60.50

**HELP**
**Remember** GST in a price (GST inc.) = price ÷ 11

**3** A shirt costs $56. It is **discounted by 26%**. What is the new price? **1 mark**

(A) $14.56 (B) $26.56 (C) $30.00 (D) $41.44

**HELP**
**Remember** Find 26% of $56, then **take it off** the price.

**4** Find the **simple interest** on $600 at 5% per year for 4 years. **1 mark**

(A) $12.00 (B) $30.00 (C) $120.00 (D) $720.00

**HELP**
**Formula** $I = P \times R \times N$ (5% → 0.05)

**5** Use a multiplier and your calculator. **2 marks**

**a** Increase $968 by 12%.

Working

Answer $ 1 084.16

**b** Decrease $4 900 by 7%.

Working

Answer $ 4 557.00

**HELP**
**Remember** Increase by 8% → × 1.08
Decrease by 8% → × 0.92
**Example** Increase $50 by 10%:
50 × 1.10 = $55

**6** A shop buys a cap for \$7.60. This is the **cost price**. The shop adds a **50% mark-up**. **2 marks**

**a** Find the selling price of the cap.

Working

**Answer** \$ **11.40**

**b** How much profit does the shop make?

Working

**Answer** \$ **3.80**

**HELP**

**Remember** Mark-up of 50% → × 1.5
Profit = selling price − cost price

**Example** Cost \$10, mark-up 20%: 10 × 1.2 = \$12, profit = \$2

**7** Tom is paid \$15.78 per hour. **2 marks**

**a** Tom works 30 hours. How much is he paid?

Working

**Answer** \$ **473.40**

**b** Tom also works 4 hours of overtime at **time and a half**. How much is he paid for the overtime?

Working

**Answer** \$ **94.68**

**HELP**

**Remember** Pay = hours × hourly rate
Time and a half = 1.5 × hourly rate

**Example** \$20 per hour, 3 hours at time and a half: 3 × 1.5 × 20 = \$90

**8** Pedro sells phones. He earns **4.5% commission** on his sales. His sales this week are \$790. How much commission does he earn? **1 mark**

Working

**Answer** \$ **35.55**

**HELP**

**Remember** 4.5% → 0.045
Commission = 0.045 × sales

**9** Tranh works 36 hours at $28.89 per hour. **2 marks**

**a** Find Tranh's **gross pay** (total pay before anything is taken out).

Working

**Answer** $ **1 040.04**

**HELP**
**Remember** Gross pay = hours × hourly rate
Net pay = gross pay − tax − super

**b** Tranh pays $142.59 in tax and $22.50 in super. Find his **net pay**.

Working

**Answer** $ **874.95**

**10** $5 000 is invested at 4% per year, **compounding** each year for 3 years. **2 marks**

**a** Find the value of the investment after 3 years.

Working

**Answer** $ **5 624.32**

**HELP**
**Formula** $A = P(1 + R)^n$
$P$ = amount invested, $R$ = rate as a decimal, $n$ = number of years
**Example** $2 000 at 5% for 2 years:
$A = 2000 \times 1.05^2 = \$2\,205$
Interest = 2205 − 2000 = $205
**Calculator** Type 2000 × 1.05 ^ 2 =

**b** How much interest is earned?

Working

**Answer** $ **624.32**

SECTION B

# Measurement

Questions **11–20** · **15 marks** · Suggested time **20 minutes**

____ / 15
Marker

Questions 11–14 are **multiple choice**. Circle the letter of the correct answer, like this: (B)

**11** How many metres are in 1 kilometre? **1 mark**

(A) 10 (B) 100 **(C) 1 000** (D) 10 000

**12** What is the **area** of a rectangle 3 cm long and 4 cm wide? **1 mark**

(A) 7 cm² (B) 12 cm (C) 14 cm **(D) 12 cm²**

**13** What is the **volume** of a cube with sides 3 cm long? **1 mark**

(A) 9 cm³ **(B) 27 cm³** (C) 27 cm² (D) 54 cm³

**HELP**
**Formula** Volume of a cube = side × side × side

**14** Which of these is 232 000 written in **scientific notation**? **1 mark**

**(A)** $2.32 \times 10^5$ (B) $23.2 \times 10^4$
(C) $2.32 \times 10^{-5}$ (D) $232 \times 10^3$

**HELP**
**Example** $2\,350\,000 = 2.35 \times 10^6$ (one digit before the decimal point)

**15** Convert each measurement. **2 marks**

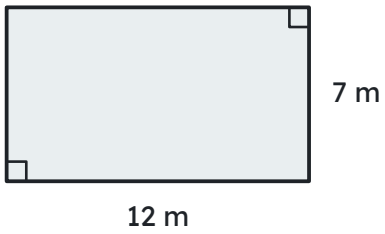
**a** Change 5.3 km into metres. **Answer** **5 300**

**b** Change 2.5 L into millilitres. **Answer** **2 500**

**HELP**
**Remember** km → m: × 1000
L → mL: × 1000

**16** A farmer puts a fence all the way around this rectangular paddock. How many metres of fence are needed? **1 mark**

**HELP**
**Remember** Perimeter = add up the lengths of **all** the sides.

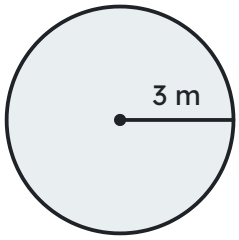



Working

**Answer** 38 m

**17** This circle has a radius of 3 m. Give your answers to 1 decimal place. **2 marks**

**a** Find the **circumference**.




Working

**Answer** 18.8 m

**b** Find the **area**.

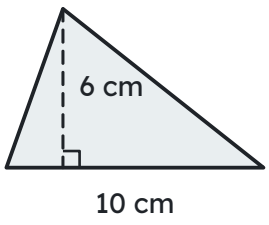
Working

**Answer** 28.3 m²

**HELP**
**Formula** $C = 2 \times \pi \times r$ (circumference)
$A = \pi \times r \times r$ (area)
**Example** Radius 5 cm:
$C = 2 \times \pi \times 5 = 31.4$ cm
$A = \pi \times 5 \times 5 = 78.5$ cm²

**18** Find the area of each shape. **2 marks**

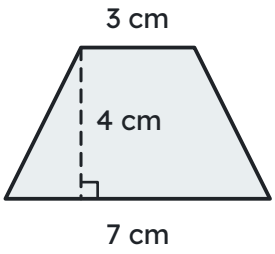
**a** Triangle




Working

**Answer** 30 cm²

**b** Trapezium

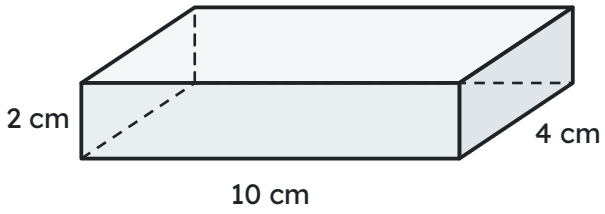



Working

**Answer** 20 cm²

**HELP**
**Formula** Triangle: $A = \frac{1}{2} \times$ base × height
Trapezium: $A = \frac{1}{2} \times (a + b) \times h$
($a$ and $b$ are the parallel sides)
**Example** Triangle, base 8 cm, height 5 cm:
$\frac{1}{2} \times 8 \times 5 = 20$ cm²

**19** Find the **surface area** of this box. Complete the table. The first row is done for you. **2 marks**



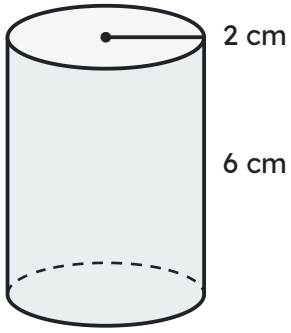

**HELP**
**Remember** A box has 6 faces, in 3 matching pairs. Surface area = add the areas of all 6 faces.

| Faces | Area of one face | Number of faces | Area |
|---|---|---|---|
| **Top and bottom** | 10 × 4 = 40 cm² | 2 | 80 cm² |
| **Front and back** | **10 × 2 = 20 cm²** | 2 | **40 cm²** |
| **The two ends** | **4 × 2 = 8 cm²** | 2 | **16 cm²** |
| **Surface area (add the last column)** | | | **136 cm²** |

**20** This cylinder has a radius of 2 cm and a height of 6 cm. **2 marks**

**a** Find the **volume**, to 1 decimal place.

Working




**Answer** 75.4 cm³

**b** How many millilitres (mL) of water can it hold?

Working

**HELP**
**Formula** Volume of a cylinder: $V = \pi \times r \times r \times h$
**Example** $r = 3$ cm, $h = 10$ cm: $V = \pi \times 3 \times 3 \times 10 = 282.7$ cm³
**Remember** 1 cm³ holds 1 mL

**Answer** 75.4 mL

SECTION C

# Statistics and probability

Questions **21–28** · **14 marks** · Suggested time **20 minutes**

_____ / 14
Marker

Questions 21–24 are **multiple choice**. Circle the letter of the correct answer, like this: (B)

**21** A survey asks: **"What is your favourite sport to play?"** What type of data does this give? **1 mark**

(A) numerical and discrete
(B) numerical and continuous
(C) categorical and ordinal
(D) categorical and nominal

**HELP**
**Remember** Categorical = words or groups. Numerical = numbers. Nominal = no order (colours, sports). Ordinal = has an order (small, medium, large).

Use this **stem-and-leaf plot** for Questions **22** and **23**. It shows the scores of 11 players in a game.

| Stem | Leaf |
|---|---|
| 2 | 4 9 |
| 3 | 1 1 7 8 |
| 4 | 2 4 6 |
| 5 | 0 4 |

4 | 2 means 42

**22** What is the **lowest** score? **1 mark**

(A) 4 (B) 20 (C) 24 (D) 54

**23** What is the **mode** (the most common score)? **1 mark**

(A) 3 (B) 31 (C) 38 (D) 42

**24** A normal 6-sided die is rolled once. What is the probability of rolling an **even** number? **1 mark**

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$

**HELP**
**Formula** Probability = number of ways it can happen ÷ total number of outcomes

**25** Here are the number of goals scored by a netball player in 7 games: **3 marks**

**8, 3, 13, 1, 5, 3, 9**

Write the numbers in order first: 1 3 3 5 8 9 13

**a** Find the **mean**. Answer 6

**b** Find the **median**. Answer 5

**c** Find the **range**. Answer 12

**HELP**

**Mean** Add up the numbers, then ÷ how many there are

**Median** The middle number, when they are in order

**Range** Biggest – smallest

**26** One letter is chosen at random from the word **2 marks**

I N T E R E S T

**a** Find the probability that the letter is an **E**.

Answer $\frac{2}{8}$ or $\frac{1}{4}$

**b** Find the probability that the letter is a **vowel** (A, E, I, O or U).

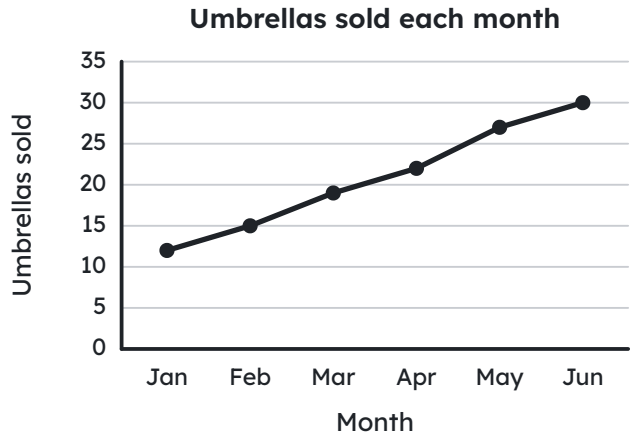
Answer $\frac{3}{8}$

**HELP**

**Formula** Probability = number of ways it can happen ÷ total number of outcomes

**Example** A letter from **BOOK**: P(O) = $\frac{2}{4} = \frac{1}{2}$

**27** This **time-series graph** shows how many umbrellas a shop sold each month. **1 mark**




**HELP**

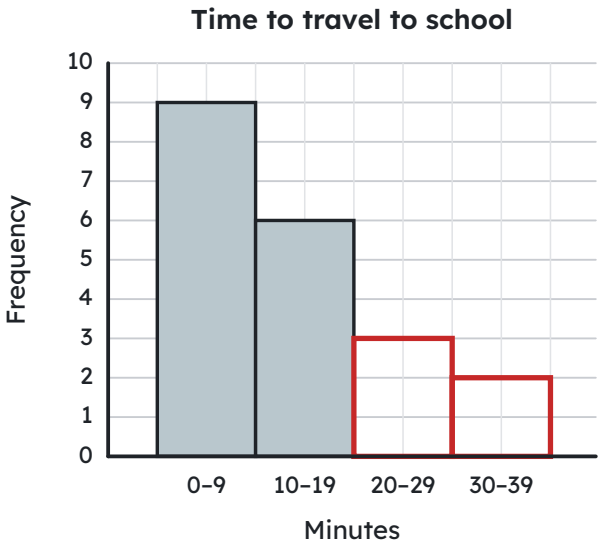
**Remember** Upward trend: the line goes up over time.
Downward trend: the line goes down over time.

Describe the **trend**. Circle one.

**upward trend** | downward trend | no trend

**28** Some students recorded how many minutes it takes them to travel to school. **4 marks**

| Minutes | Frequency |
|---|---|
| 0–9 | 9 |
| 10–19 | 6 |
| 20–29 | 3 |
| 30–39 | 2 |
| **Total** | **20** |




**a** Fill in the **total** at the bottom of the table.

Total = 9 + 6 + 3 + 2 = 20

**b** What **percentage** of the students took 0–9 minutes?

Working

**HELP**

**Formula** Percentage = frequency ÷ total × 100

**Shapes**




Answer 45 %

**c** Two columns of the histogram are drawn. **Draw the columns** for 20–29 and 30–39.

Columns of height 3 and 2, touching, shown on the graph.

**d** Is the data **symmetrical** or **skewed**? Circle one.

symmetrical **skewed**

SECTION D

# Other Year 10 topics

Questions **29–34** · **11 marks** · Suggested time **15 minutes**

_____ / 11
Marker

Question 29 is **multiple choice**. Circle the letter of the correct answer, like this: (B)

**29** Simplify $4x + 3x$ — **1 mark**

(A) $7x$ (B) $12x$ (C) $7x^2$ (D) $7$

**30** If $a = 4$, find the value of $5a - 2$. — **1 mark**

Working

**HELP**
**Remember** $5a$ means $5 \times a$
**Example** If $a = 3$, then $2a + 1 = 2 \times 3 + 1 = 7$

**Answer** 18

**31** Solve each equation. — **2 marks**

**a** $x + 9 = 15$

Working

**HELP**
**Remember** Do the **opposite** to both sides.
**Example** $3x + 1 = 13$
Take away 1: $3x = 12$
Divide by 3: $x = 4$

**Answer** $x =$ 6

**b** $2x + 3 = 13$

Working

**Answer** $x =$ 5

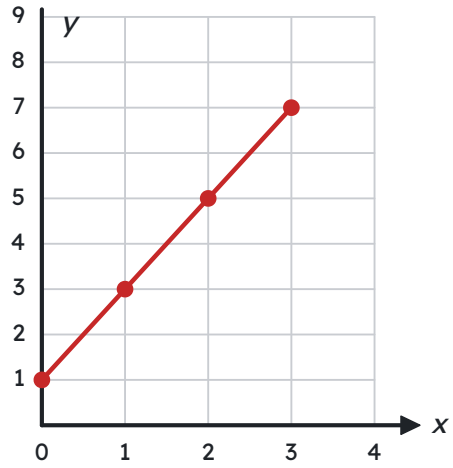
**32** The rule for a straight line is $y = 2x + 1$. **3 marks**

**a** Complete the table of values.

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $y$ | **1** | **3** | **5** | **7** |

**2 marks: all four correct. 1 mark: two or three correct.**

**b** Plot the points on the grid and join them with a straight line.



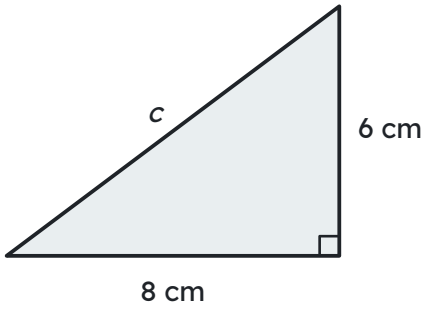

**Points plotted correctly and joined with a ruled straight line.**

**HELP**

**Example** For $y = 3x + 2$, when $x = 4$:
$y = 3 \times 4 + 2 = 14$

**33** Find the length of the **hypotenuse**, $c$. **2 marks**

Working




**HELP**

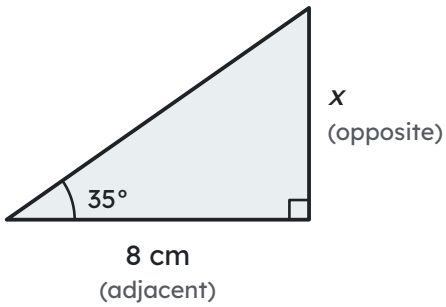
**Formula** $c^2 = a^2 + b^2$ ($c$ is the longest side)

**Example** $a = 3$, $b = 4$:
$c^2 = 9 + 16 = 25$
$c = \sqrt{25} = 5$

**Answer** $c =$ **10** cm

**34** Find the length of side $x$, to 1 decimal place. **2 marks**

Working




**HELP**

**Formula** tan θ = opposite ÷ adjacent
so opposite = adjacent × tan θ

**Example** Angle 20°, adjacent 5 cm:
$x = 5 \times \tan 20° = 1.8$ cm

**Calculator** Check your calculator shows **D** or **DEG** (degrees).

**Answer** $x =$ **5.6** cm

## End of the test

- Did you answer **every** question? Have a go at any you missed.
- Did you write the **units** ($, cm, m², mL) and show your **working**?
- Use the space below if you need more room. Write the question number next to your working.

Extra working space